

# Randomized Algorithms

## Random Walks on Undirected Graphs

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# Outline

- 1 The random-walk model
- 2 Stationary distribution
- 3 Hitting and commute times
- 4 Cover time
- 5 Application:  $s$ - $t$  connectivity



# Why random walks on graphs?

- A random walk is a Markov chain whose states are vertices of a graph.
- It is a simple model of local exploration: from the current vertex, choose a random neighbor.
- In this lecture note, we use graph structure to bound:
  - stationary distributions,
  - hitting and commute times,
  - cover times,
  - and a randomized low-space algorithm for  $s$ - $t$  connectivity.

## Guiding question

How long does local random exploration need to see a target vertex, or even all vertices?



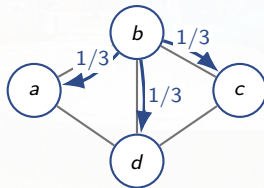
# Random walk on an undirected graph

Let  $G = (V, E)$  be a finite undirected graph.

## Definition 1

At vertex  $u$ , the standard random walk moves to a uniformly chosen neighbor:

$$P_{u,v} = \begin{cases} 1/d(u), & (u,v) \in E, \\ 0, & \text{otherwise.} \end{cases}$$



## The transition matrix is local

- The transition matrix  $P$  has one row per vertex.
- Row  $u$  has nonzero entries only at neighbors of  $u$ .
- The degree  $d(u)$  controls the transition probabilities leaving  $u$ .

### Example

If  $N(u) = \{v_1, v_2, v_3\}$ , then

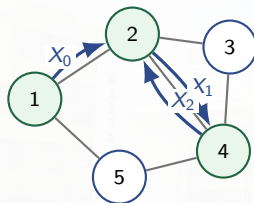
$$P_{u,v_1} = P_{u,v_2} = P_{u,v_3} = \frac{1}{3}.$$

### Memoryless property

Given the current vertex, the previous trajectory carries no additional information for choosing the next vertex.



# A sample walk



A walk is a random sequence

$$X_0, X_1, X_2, \dots$$

with

$$\Pr[X_{t+1} = v \mid X_t = u] = P_{u,v}.$$

## Note

The graph is fixed; the randomness is in the walk.

# Structural assumptions

In the following discussions, we assume  $G$  is:

- finite,
- undirected,
- connected,
- non-bipartite.

Why?



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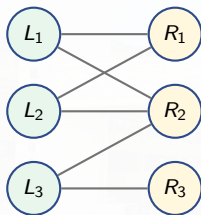
## Why?

- Connected  $\Rightarrow$  every vertex can be reached from every other vertex.
- Non-bipartite  $\Rightarrow$  the walk is aperiodic.
- Together, the *finite* Markov chain converges to a stationary distribution.





## Bipartite graphs cause period two



bipartite: alternate sides

In a bipartite graph, every step switches side.

$$L \rightarrow R \rightarrow L \rightarrow R \rightarrow \dots$$

So a return to the starting vertex can happen only at even times.

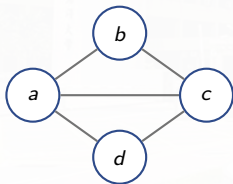
### Consequence

The period is 2, so the chain is not aperiodic.

# Non-bipartite means aperiodic

## Lemma 1

A random walk on an undirected graph  $G$  is aperiodic if and only if  $G$  is not bipartite.



odd cycle present

- Every undirected edge gives a return of length 2:  $v \rightarrow u \rightarrow v$ .
- An odd cycle gives an odd return length.
- Having return lengths with gcd 1 means period 1 (i.e., *aperiodic*).

# The stationary distribution

## Theorem 1

For a finite, connected, non-bipartite undirected graph,

$$\pi_v = \frac{d(v)}{2|E|}$$

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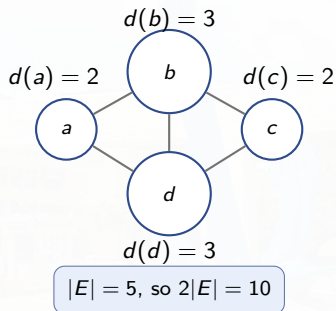
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The walk spends more time at high-degree vertices, because they are incident to more directed edge-ends.

The “distribution” is valid:  $\sum_{v \in V} \pi_v = \sum_{v \in V} \frac{d(v)}{2|E|} = 1.$



# Stationary distribution: a concrete picture



$$\pi_a = \pi_c = \frac{2}{10}, \quad \pi_b = \pi_d = \frac{3}{10}.$$

## Why the formula works: detailed balance

For every edge  $(u, v) \in E$ ,

$$\pi_u P_{u,v} = \frac{d(u)}{2|E|} \cdot \frac{1}{d(u)} = \frac{1}{2|E|}.$$

Similarly,

$$\pi_v P_{v,u} = \frac{1}{2|E|}.$$

Therefore

$$\pi_u P_{u,v} = \pi_v P_{v,u}.$$

### Meaning

In stationarity, the probability flow across each undirected edge is balanced in both directions.



# Hitting time, commute time, cover time

## Definition 2

For vertices  $u, v \in V$ , let

$$h_{u,v} = \mathbb{E}[\text{time to first reach } v \mid X_0 = u].$$

This is the *hitting time* from  $u$  to  $v$ .

## Two related quantities

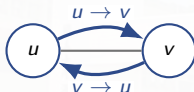
- **Commute time:**

$$h_{u,v} + h_{v,u}.$$

- **Cover time:** maximum expected time, over starting vertices, to visit every vertex.



## Directed-edge viewpoint



Replace each undirected edge by two directed edges.

- There are  $2|E|$  directed edges.
- The edge traversed at time  $t$  is a Markov-chain state.
- In stationarity, each directed edge has probability  $1/(2|E|)$ .

## Lemma 7.14: adjacent commute time

### Lemma 2

If  $(u, v) \in E$ , then

$$h_{u,v} + h_{v,u} \leq 2|E|.$$



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### Core idea

The expected return time to a directed edge  $u \rightarrow v$  is

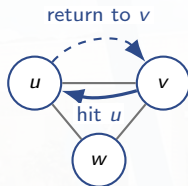
$$\frac{1}{1/(2|E|)} = 2|E|.$$

Returning to the directed edge  $u \rightarrow v$  requires going from  $v$  to  $u$  and then back to  $v$  through that edge.



## Why this bounds the whole commute

Suppose the walk has just traversed  $u \rightarrow v$ , so it is currently at  $v$ .



specific return via  $u \rightarrow v$  is only one possible way

$$h_{v,u} + h_{u,v} \leq \mathbb{E}[\text{additional time until next traversal of } u \rightarrow v] = 2|E|.$$

# A useful caution

## Why the sum appears

The return time to a directed edge controls a complete round trip:

$$V \rightsquigarrow U \rightsquigarrow V.$$

The first traversal has already occurred; the bound concerns the additional time until the next one.



# Cover time

## Definition 3

The cover time  $C_G$  is

$$C_G = \max_{v \in V} \mathbb{E}[\text{time to visit all vertices} \mid X_0 = v].$$

## Goal

Bound  $C_G$  using simpler quantities such as:

- number of edges  $|E|$ ,
- number of vertices  $|V|$ ,
- maximum hitting time  $\max_{u \neq v} h_{u,v}$ .



# Spanning-tree tour bound

## Lemma 3

For a connected undirected graph  $G = (V, E)$ ,

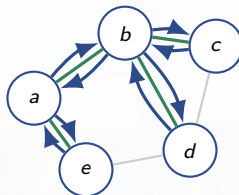
$$C_G \leq 2|E|(|V| - 1).$$

- Choose any spanning tree  $T$  of  $G$ .
- A depth-first tour crosses each tree edge once in each direction.
- Each tree-edge commute costs at most  $2|E|$  by Lemma 2.
- There are  $|V| - 1$  tree edges.





# Spanning tree tour picture



DFS tour of spanning tree: each tree edge twice

$$\sum_{(x,y) \in T} (h_{x,y} + h_{y,x}) \leq 2|E|(|V| - 1).$$

# Matthews' theorem

## Lemma 4

Let  $G$  have  $n$  vertices. Then

$$C_G \leq H(n-1) \max_{u,v \in V: u \neq v} h_{u,v},$$

where

$$H(n-1) = \sum_{i=1}^{n-1} \frac{1}{i}.$$

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Why and when is it useful?

If all pairwise hitting times are small, the cover time is like coupon collector with hitting-time cost.



# Matthews' theorem: proof idea (1/3)

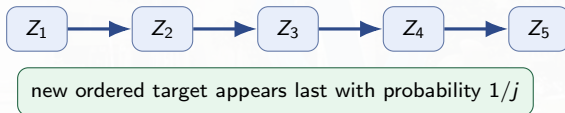
Let

$$B = \max_{u \neq v} h_{u,v}.$$

Choose a uniformly random ordering of the vertices:

$$Z_1, Z_2, \dots, Z_n.$$

Let  $T_j$  be the first time by which all of  $Z_1, Z_2, \dots, Z_j$  have been visited.



For  $j \geq 2$ , the extra waiting interval satisfies

$$\mathbb{E}[T_j - T_{j-1}] \leq \frac{B}{j}. \quad (\text{Why?})$$

Summing gives a harmonic factor.



## Matthews' theorem: proof idea (2/3)

Say the chain's history is given by  $X_1, X_2, \dots$ . For  $j \geq 2$ , we consider

$$Y_j = \mathbb{E}[T_j - T_{j-1} \mid Z_1, Z_2, \dots, Z_j; X_1, X_2, \dots, X_{T_{j-1}}].$$

- **Note:** During the random walk, these vertices  $Z_1, Z_2, \dots$  are not necessarily visited in this same order.



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The expected number to cover the graph starting from vertex  $u$ :

$$\sum_{j=2} Y_j + \mathbb{E}[T_1].$$



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- If  $Z_1 = u$ , which happens w.p.  $\frac{1}{n}$ , then  $T_1 = 0$ .
- Otherwise,  $\mathbb{E}[T_1 \mid Z_1] = h_{u, Z_1} \leq B$ .
- Hence,  $\mathbb{E}[T_1] = (1 - \frac{1}{n})B$ .



## Matthews' theorem: proof idea (3/3)

For the  $Y_i$ :

- If  $Z_j$  is not the last vertex seen from  $Z_1, Z_2, \dots, Z_j$ , then  $Y_i = 0$  since  $T_j = T_{j-1}$ .
- If  $Z_j$  is the last vertex seen from  $Z_1, Z_2, \dots, Z_j$ , then  $Y_j \leq B$ .
  - $Y_j$ : the hitting time  $h_{Z_k, Z_j}$  (say  $Z_k$  was visited last out of  $Z_1, Z_2, \dots, Z_{j-1}$  during the random walk).
  - $Z_j$  was chosen according to a random permutation, independent of the random walk, so  $Z_j$  is last out of  $Z_1, Z_2, \dots, Z_{j-1}$  with probability  $1/j$ .





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$$\begin{aligned} \sum_{j=2}^n Y_j + \mathbb{E}[T_1] &\leq \sum_{j=2}^n \frac{1}{j} B + \left(1 - \frac{1}{n}\right) B = \left(1 + \sum_{j=2}^n \frac{1}{j}\right) B - \frac{1}{n} B \\ &= H(n-1)B. \end{aligned}$$



# Lemma 3 vs. Lemma 4

|              | Lemma 3              | Lemma 4                            |
|--------------|----------------------|------------------------------------|
| Input needed | $ E $ and $ V $      | maximum hitting time               |
| Bound        | $2 E ( V  - 1)$      | $H(n - 1) \max_{u \neq v} h_{u,v}$ |
| Method       | spanning-tree tour   | random ordering / coupon collector |
| Best when    | no hitting-time info | good hitting-time estimates exist  |

## Note

Lemma 3: coarse but always available.

Lemma 4: can be sharper when the graph mixes or spreads quickly.



## $s$ - $t$ connectivity with tiny memory

Given an undirected graph  $G = (V, E)$  and two vertices  $s, t$ , decide whether  $s$  and  $t$  are connected.

- Breadth-first search or depth-first search solves this in linear time.
- But such **deterministic** graph search needs  $\Theta(n)$  space to **remember visited vertices**.
  - Can be done in  $O(n + m)$  time by BFS/DFS.
- The random-walk algorithm **uses only the current vertex** and a **step counter**.

### Goal

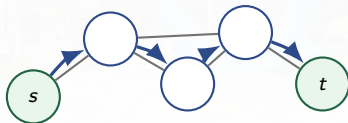
Trade time and error probability for very **small space**:  $O(\log n)$  bits.



# Algorithm 1: Testing connectivity by random-walks

## $s$ - $t$ connectivity algorithm

- 1 Start a random walk from  $s$ .
- 2 If the walk reaches  $t$  within  $2n^3$  steps, return “connected.”
- 3 Otherwise return “not connected.”



## Correctness and one-sided error

- If there is no path from  $s$  to  $t$ , the walk can never reach  $t$ .
- Therefore returning “connected” is always correct.
- The only possible error is saying “not connected” even though a path exists.

### Bounding the error

If  $s$  and  $t$  are connected, the expected time to reach  $t$  from  $s$  is at most the cover time of their component.

$$\mathbb{E}[T_{s \rightarrow t}] \leq C_G \leq 2nm < n^3.$$

Then Markov's inequality gives

$$\Pr[T_{s \rightarrow t} > 2n^3] \leq \frac{1}{2}.$$

# Space usage and amplification

- Store current vertex:  $O(\log n)$  bits.
- Store step counter up to  $2n^3$ :  $O(\log n)$  bits.
- Choose a random neighbor at each step.

## Amplification

Repeat the random walk independently  $r$  times.

$$\Pr[\text{"false negative after } r \text{ runs"}] \leq 2^{-r}.$$

The space remains  $O(\log n)$ ; the time grows by a factor of  $r$ .



# Takeaways (1/2)

- Random walks on connected non-bipartite undirected graphs converge to

$$\pi_v = \frac{d(v)}{2|E|}.$$

- Directed-edge stationarity gives the adjacent commute-time bound

$$h_{u,v} + h_{v,u} \leq 2|E|.$$

- Cover time can be bounded by either spanning-tree tours or Matthews' theorem.



## Takeaways (2/2)

- The random-walk  $s$ - $t$  connectivity algorithm is simple, one-sided, and uses only  $O(\log n)$  space.
- In fact, Reingold's theorem later showed that even the randomization can be removed. [Omer Reingold J. ACM 2008]





# Discussions

